NPRB2301 Sample and Data Update

November 5, 2025

This report summarizes the samples available for NPRB 2301, their status and the status of laboratory processes. Updates will be made available whenever significant updates to the sample database or analytical data become available.

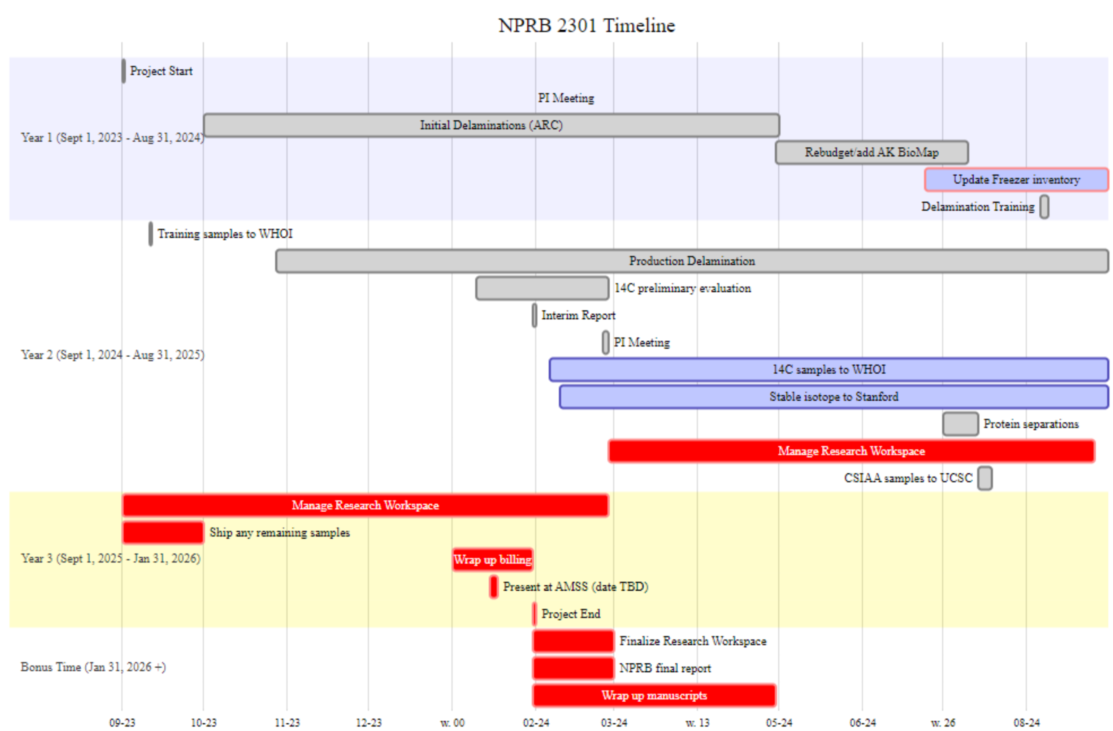
Version Notes

- Updated diameter and sample submission information (NPRB2301_layer_results), NOTE: the updated diameters are for samples that have not been submitted for 14C, SIA or CSIAA yet, thus will not be reflected in the NPRB2301_layer_results
- Updated large batch of SIA results, there are still 68 NOSAMS samples and 20 CSIA-AA samples pending analysis
- Fixed the "samples remaining in the budget" to reflect the increased rate at NOSAMS, from $268/sample to $276/sample. This rate change reduces our final sample size down to about 500 samples.

Links and Resources

- github repo (public)
- Research Workspace (password protected)
- Google Folder (access controlled)

Project Timeline (not updated)



Specimen, Sample and Layer Summaries

- Each animal has a specimen_ID. The exception is spiny dogfish embryos, which are considered samples taken of the adult.
- Each eye, embryo, spine, etc. has a unique sample_id, which links back to the specimen_id that that sample was taken from.

- Each eye is further separated into layers and each layer_id links back to the sample_id (and thus, specimen_id).Layers are numbered sequentially from the outside towards the center. Early numbering convention used “N” for nucleus, and “R” for the remainder.
- Embryo each have a sample_id, but two eyes. Each eye and/or layer is labeled with the sample_id, L or R and layer number (as applicable).

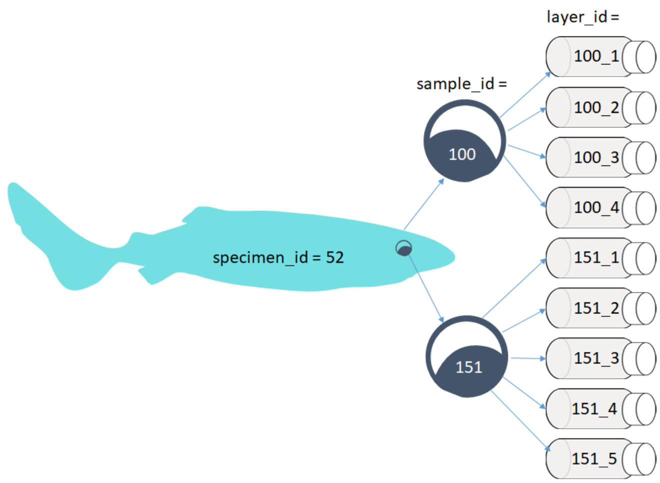


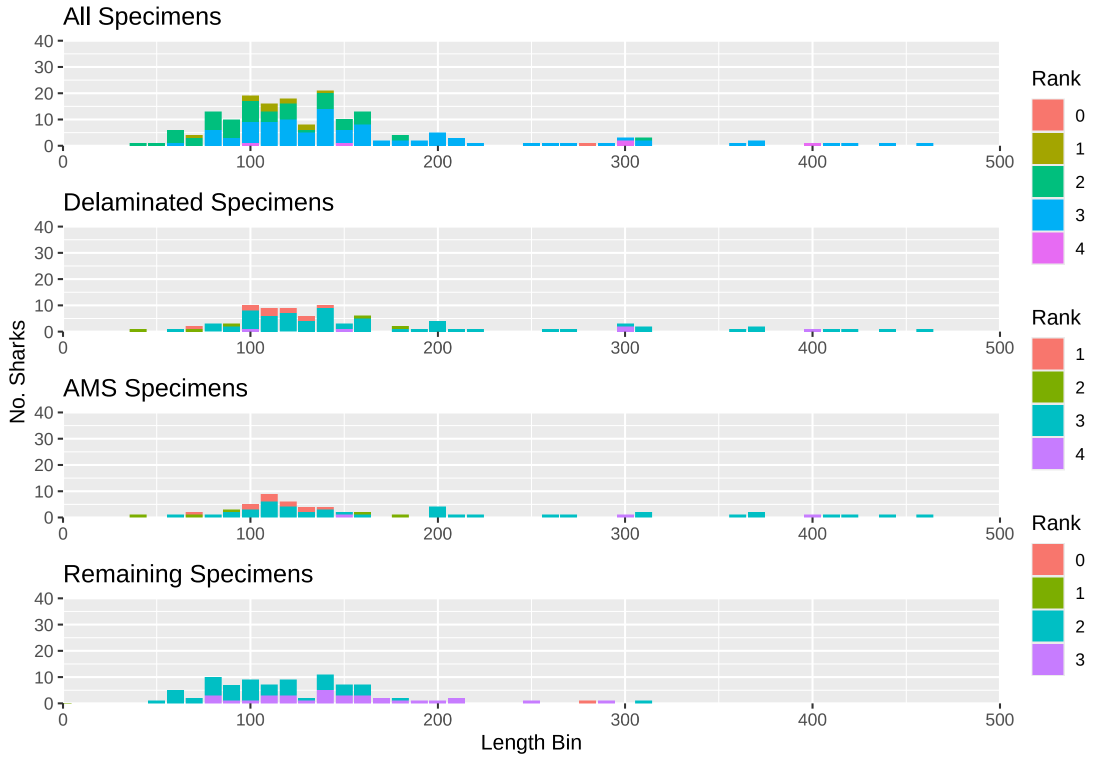
Figure 1: Specimens to samples to layers flow chart

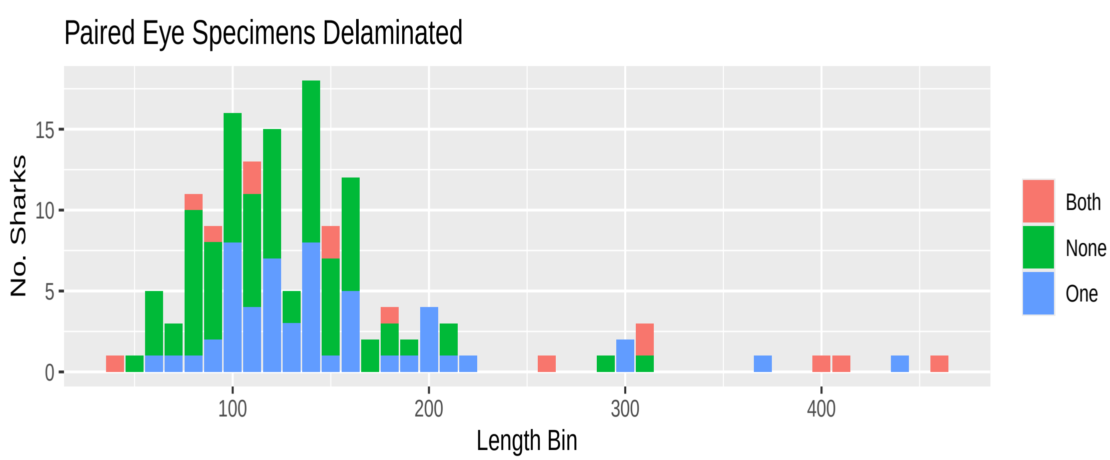
Pacific sleeper sharks

The below table only includes specimens with both length and haul data. There are some samples without haul data, and some without length. As data gaps get filled in, these numbers will change. Please request any other data breakdowns for future reports.

N PSS	N PSS eyes	N PSS Delaminated	N PSS Layers
173	320	113	678

For the purposes of the below graphs, all unknown length types are treated as Total Length until we resolve how to determine that. Ranks are from 1 (lowest) to 4 (highest) and described here. Rank = 1 are samples that had poor handling and may have thawed before delamination, Rank = 2 are collections with missing length or haul data, but do have at least a “source” (being who provided the sample), Rank = 3 are collections with complete data, but some uncertainties (i.e., length type), and Rank = 4 are survey collections with best chain of custody and full data available. Many specimens have not yet been attributed to a “source” in the database, this is intentional. Many of the at-sea observer provided samples did not have full data with the sample, or deck forms submitted. Some of these data gaps will be filled as we dig through at-sea observer databases, and those specimens will change from rank = 0 to the appropriate rank 1 - rank 4 value.





Spiny dogfish

N SD	N SD with full data	N SD available	N SD with samples but no length
28	17	12	5

This section has not been updated, pending new samples arriving from BC and Washington. Stay tuned.

Carbon 14

The estimated N 14C samples left per budget reflects those that have been billed, not all of the samples that have been submitted.

N 14C sent	N 14C returned	N 14C samples left per budget	Total N budgeted
497	466	1	498

The below table summarizes the number of each species which have 14C results from at least one layer

N 14C PSS	N 14C SD
61	9

14C results can be found in NPRB2301_layer_results.

Date Sent	N layers sent	N layers returned
Mar 25, 2019	11	11
Aug 5, 2019	8	8
Sept 11, 2024	36	35
Feb 6, 2025	70	70
Mar 31, 2025	96	95
Apr 28, 2025	50	50
July 14, 2025	68	0

Stable Isotopes

N SIA sent	N SIA returned	N SIA samples left per budget	Total N Budgeted
506	295	265	560

Date Sent	N layers sent	N layers returned
Feb 4, 2025	70	70
Mar 31, 2025	107	107
Apr 1, 2025	15	15
Apr 25, 2025	35	35
July 14, 2025	68	68

Stable isotope results can be found in NPRB2301_layer_results.

CSIA-AA

NOTE: we have funds for more samples than initially planned. Total sample size is now 30. We will proceed with the plan for the first 20, then discuss how to use the remaining samples. Plan: sequence layers (n=5) from either 507 or 508 (which ever has least layers for AMS) sequence layers (n=5) from 678 core and outer layer (n=2) from 161, 273, 422, 473, and 544

N CSIA_AA sent	N CSIA_AA returned	N CSIA_AA samples left per budget	Total N Budgeted
29	0	30	30

Protein Separation

Protein separation is proceeding as planned. Sleeper shark samples are complete and Charlotte is working on the dogfish samples sent by WA.

Spiny Dogfish Ageing

Cindy, Beth and Charlotte have concluded preliminary reads of all of the spiny dogfish spines following protocols in Tribuzio et al. 2017. Those results are available in the NPRB2301_dogfish_spine_ages_PRELIMINARY. The age estimates follow the estimation methods recommended by Tribuzio et al. 2010.

The spines are being sent to the WDFW ageing lab, as they are the current authority on ageing spiny dogfish and final age estimates will be provided once WDFW completes their reads.